Modelling and control of multibody mechanical systems

Model answers

Question 1

a) One single-axis-rotation transformation matrix can be used, i.e.

$$\boldsymbol{C_\theta} = \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix},$$

which is the rotation matrix by angle θ about a y axis. The body angular velocity vector in the body-fixed coordinate system is given by:

$$\boldsymbol{\Omega} = \boldsymbol{C_\theta} \begin{bmatrix} 0 \\ 0 \\ \dot{\psi} \end{bmatrix} + \begin{bmatrix} 0 \\ \dot{\theta} \\ 0 \end{bmatrix} = \begin{bmatrix} -\dot{\psi}\sin\theta \\ \dot{\theta} \\ \dot{\psi}\cos\theta \end{bmatrix}. \tag{1}$$

[5 marks]

b) The kinetic energy of the compound pendulum is the kinetic energy of the rigid body for its rotational motion about the fixed point, that is:

$$T = \frac{1}{2} I_{xx} \dot{\psi}^2 \sin^2\theta + \frac{1}{2} I_{xx} \dot{\theta}^2 + \frac{1}{2} I_{zz} \dot{\psi}^2 \cos^2\theta.$$

[5 marks]

Answers

c) The potential energy of the compound pendulum is changing because the centre of mass of the rigid body is changing height. Therefore, the potential energy of the compound pendulum is

$$V = -mgf\cos\theta,$$

with the point O the zero potential energy level. Hence the Lagrangian is $L = T - V$, which gives

$$L = \frac{1}{2} I_{xx} \dot{\psi}^2 \sin^2\theta + \frac{1}{2} I_{xx} \dot{\theta}^2 + \frac{1}{2} I_{zz} \dot{\psi}^2 \cos^2\theta + mgf\cos\theta.$$

[4 marks]

d) The Lagrangian equation with respect to the generalised coordinate ψ is

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{\psi}}\right) - \frac{\partial L}{\partial \psi} = 0,$$

which gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(I_{xx}\dot{\psi}\sin^2\theta + I_{zz}\dot{\psi}\cos^2\theta\right) = 0.$$

The Lagrangian equation with respect to the generalised coordinate θ is

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{\theta}}\right) - \frac{\partial L}{\partial \theta} = 0,$$

which gives

$$I_{xx}\ddot{\theta} + (I_{zz} - I_{xx})\dot{\psi}^2 \sin\theta\cos\theta + mgf\sin\theta = 0.$$

[6 marks]

Question 2

a) The position vector of the centre of mass of rod OD is

$$\boldsymbol{r_{C_1}} = h_1 \boldsymbol{e_\theta}$$

and the position vector of the centre of mass of rod AB is

$$\boldsymbol{r_{C_2}} = h_2 \boldsymbol{e_\theta}.$$

The velocity vectors are found by differentiating the corresponding position vectors. The velocity vector of the centre of mass of rod OD is

$$\dot{\boldsymbol{r}}_{\boldsymbol{C_1}} = -h_1 \dot{\theta} \boldsymbol{e_r}$$

and the velocity vector of the centre of mass of rod AB is

$$\dot{\boldsymbol{r}}_{\boldsymbol{C_2}} = -h_2 \dot{\theta} \boldsymbol{e_r}$$

[4 marks]

b) The total kinetic energy of the system is the sum of the kinetic energies of the two rods, that is:

$$\begin{aligned} T &= \frac{1}{2} m_1 \dot{\boldsymbol{r}}_{\boldsymbol{C_1}} \cdot \dot{\boldsymbol{r}}_{\boldsymbol{C_1}} + \frac{1}{2} I_1 \dot{\theta}^2 + \frac{1}{2} m_2 \dot{\boldsymbol{r}}_{\boldsymbol{C_2}} \cdot \dot{\boldsymbol{r}}_{\boldsymbol{C_2}} + \frac{1}{2} I_2 \left(\dot{\theta} + \dot{\phi}\right)^2 \\ &= \frac{1}{2} m_1 h_1^2 \dot{\theta}^2 + \frac{1}{2} I_1 \dot{\theta}^2 + \frac{1}{2} m_2 h_2^2 \dot{\theta}^2 + \frac{1}{2} I_2 \left(\dot{\theta} + \dot{\phi}\right)^2 \\ &= \frac{1}{2} \left(m_1 h_1^2 + m_2 h_2^2 + I_1\right) \dot{\theta}^2 + \frac{1}{2} I_2 \left(\dot{\theta} + \dot{\phi}\right)^2 \end{aligned}$$

[5 marks]

Answers

c) The potential energy of both rods is changing, therefore the potential energy of the system is

$$\begin{aligned} V &= m_1 g h_1 \cos\theta + m_2 g h_2 \cos\theta \\ &= (m_1 h_1 + m_2 h_2)\, g \cos\theta, \end{aligned}$$

with the ground (point O) the zero potential energy level.
Hence the Lagrangian is $L = T - V$, which gives

$$L = \frac{1}{2} \left(m_1 h_1^2 + m_2 h_2^2 + I_1\right) \dot{\theta}^2 + \frac{1}{2} I_2 \left(\dot{\theta} + \dot{\phi}\right)^2 - (m_1 h_1 + m_2 h_2)\, g \cos\theta.$$

[4 marks]

d) The Lagrangian equation with respect to the generalised coordinate θ is

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{\theta}}\right) - \frac{\partial L}{\partial \theta} = 0,$$

which gives

$$\left(m_1 h_1^2 + m_2 h_2^2 + I_1\right) \ddot{\theta} + I_2 \left(\ddot{\theta} + \ddot{\phi}\right) - (m_1 h_1 + m_2 h_2)\, g \sin\theta = 0.$$

The Lagrangian equation with respect to the generalised coordinate ϕ is

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{\phi}}\right) - \frac{\partial L}{\partial \phi} = T_c,$$

yielding

$$I_2 \left(\ddot{\theta} + \ddot{\phi}\right) = T_c.$$

[7 marks]

Question 3

a) i) Two single-axis-rotation transformation matrices are needed, i.e.

$$\boldsymbol{D_\psi} = \begin{bmatrix} \cos\psi & \sin\psi & 0 \\ -\sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

which is the rotation matrix by an angle ψ about the z axis, and

$$\boldsymbol{C_\theta} = \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix},$$

which is the rotation matrix by an angle θ about the y axis.
The transformation from earth-fixed coordinates to plate-fixed coordinates is simply $\boldsymbol{A_1} = \boldsymbol{D_\psi}$ and the transformation from body-fixed coordinates to plate-fixed coordinates is simply $\boldsymbol{A_2} = \boldsymbol{C_\theta^T}$. [3 marks]

ii) The body angular momentum vector in body fixed coordinates is given by $\boldsymbol{H} = \boldsymbol{I\Omega}$, where $\boldsymbol{I}$ is the inertia matrix and $\boldsymbol{\Omega}$ is the angular velocity vector in the body-fixed coordinate system. To compute $\boldsymbol{H}$ in the plate-fixed coordinate system the transformation $\boldsymbol{A_2}$ is used. Hence $\boldsymbol{H}$ in plate-fixed coordinates is given by

$$\boldsymbol{H} = \boldsymbol{C_\theta^T I\Omega} =$$

$$\begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix} \begin{bmatrix} I_{xx} & 0 & 0 \\ 0 & I_{xx} & 0 \\ 0 & 0 & I_{zz} \end{bmatrix} \left(\begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ \dot\psi \end{bmatrix} + \begin{bmatrix} 0 \\ \dot\theta \\ 0 \end{bmatrix} \right) =$$

$$\begin{bmatrix} (I_{zz} - I_{xx})\dot\psi \sin\theta\cos\theta \\ I_{xx}\dot\theta \\ I_{xx}\dot\psi\sin^2\theta + I_{zz}\dot\psi\cos^2\theta \end{bmatrix}.$$

[5 marks]

Answers

b) The moment due to gravity is in the y-axis of the plate and has a magnitude of $-mgf\sin\theta$. The compound pendulum is free to rotate about the z-axis of the plate (it rotates with the plate) and about the y-axis of the plate. It is restricted from rotating about the x-axis of the plate. This constraint is maintained by a moment in the direction of this axis and of magnitude M. Therefore, the total external moment is given by

$$\boldsymbol{N} = \begin{bmatrix} M \\ -mgf\sin\theta \\ 0 \end{bmatrix}.$$

[4 marks]

c) By considering the rotational motion of the rigid body about the fixed point O, according to $\frac{d\boldsymbol{H}}{dt} = \boldsymbol{N}$ evaluated using the plate-fixed coordinate system, one has

$$\frac{\mathrm{d}'\boldsymbol{H}}{\mathrm{d}t} + \boldsymbol{\Omega}_{\text{plate}} \times \boldsymbol{H} = \boldsymbol{N},$$

where

$$\boldsymbol{\Omega}_{\text{plate}} = \begin{bmatrix} 0 \\ 0 \\ \dot{\psi} \end{bmatrix}$$

is the angular velocity vector of the plate-fixed coordinate system. Therefore,

$$\begin{bmatrix} \frac{\mathrm{d}}{\mathrm{d}t}\left((I_{zz}-I_{xx})\dot{\psi}\sin\theta\cos\theta\right) \\ I_{xx}\ddot{\theta} \\ \frac{\mathrm{d}}{\mathrm{d}t}\left(I_{xx}\dot{\psi}\sin^2\theta + I_{zz}\dot{\psi}\cos^2\theta\right) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \dot{\psi} \end{bmatrix} \times \begin{bmatrix} (I_{zz}-I_{xx})\dot{\psi}\sin\theta\cos\theta \\ I_{xx}\dot{\theta} \\ I_{xx}\dot{\psi}\sin^2\theta + I_{zz}\dot{\psi}\cos^2\theta \end{bmatrix} = \begin{bmatrix} M \\ -mgf\sin\theta \\ 0 \end{bmatrix},$$

which yields

$$\begin{bmatrix} \frac{\mathrm{d}}{\mathrm{d}t}\left((I_{zz}-I_{xx})\dot{\psi}\sin\theta\cos\theta\right) \\ I_{xx}\ddot{\theta} \\ \frac{\mathrm{d}}{\mathrm{d}t}\left(I_{xx}\dot{\psi}\sin^2\theta + I_{zz}\dot{\psi}\cos^2\theta\right) \end{bmatrix} + \begin{bmatrix} -I_{xx}\dot{\theta}\dot{\psi} \\ (I_{zz}-I_{xx})\dot{\psi}^2\sin\theta\cos\theta \\ 0 \end{bmatrix} = \begin{bmatrix} M \\ -mgf\sin\theta \\ 0 \end{bmatrix}.$$

i) The two equations of motion are therefore

$$I_{xx}\ddot{\theta} + (I_{zz}-I_{xx})\dot{\psi}^2\sin\theta\cos\theta + mgf\sin\theta = 0,$$

and

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(I_{xx}\dot{\psi}\sin^2\theta + I_{zz}\dot{\psi}\cos^2\theta\right) = 0.$$

[4 marks]

ii) The moment M is given by

$$M = \frac{\mathrm{d}}{\mathrm{d}t}\left((I_{zz}-I_{xx})\dot{\psi}\sin\theta\cos\theta\right) - I_{xx}\dot{\theta}\dot{\psi}.$$

[4 marks]

Answers

Question 4

a) The generalised coordinates are θ and ϕ. [2 marks]

b) The position vector of the centre of mass of rod OD is $\boldsymbol{r_{C_1}} = h_1 \boldsymbol{e_\theta}$ and the position vector of the centre of mass of rod AB is $\boldsymbol{r_{C_2}} = h_2 \boldsymbol{e_\theta}$. By differentiating twice each of the position vectors, the acceleration vector of rod OD is given by

$$\ddot{\boldsymbol{r}}_{\boldsymbol{C_1}} = -h_1 \ddot{\theta} \boldsymbol{e_r} - h_1 \dot{\theta}^2 \boldsymbol{e_\theta},$$

and the acceleration vector of rod AB is given by

$$\ddot{\boldsymbol{r}}_{\boldsymbol{C_2}} = -h_2 \ddot{\theta} \boldsymbol{e_r} - h_2 \dot{\theta}^2 \boldsymbol{e_\theta}.$$

[3 marks]

c) The total force on rod OD in vector form is

$$\boldsymbol{F_{total1}} = (F_{r1} - F_{r2} - m_1 g \sin\theta)\, \boldsymbol{e_r} + (F_{\theta 1} - F_{\theta 2} - m_1 g \cos\theta)\, \boldsymbol{e_\theta}.$$

The total force on rod AB in vector form is

$$\boldsymbol{F_{total2}} = (F_{r2} - m_2 g \sin\theta)\, \boldsymbol{e_r} + (F_{\theta 2} - m_2 g \cos\theta)\, \boldsymbol{e_\theta}.$$

[3 marks]

Answers

d) By considering the translational motion of the centre of mass of rod AB one has

$$(F_{r2} - m_2 g \sin\theta)\, \boldsymbol{e_r} + (F_{\theta 2} - m_2 g \cos\theta)\, \boldsymbol{e_\theta} = m_2 \left(-h_2 \ddot{\theta} \boldsymbol{e_r} - h_2 \dot{\theta}^2 \boldsymbol{e_\theta}\right),$$

therefore

$$F_{r2} = -m_2 h_2 \ddot{\theta} + m_2 g \sin\theta, \tag{2}$$

$$F_{\theta 2} = -m_2 h_2 \dot{\theta}^2 + m_2 g \cos\theta. \tag{3}$$

By considering the translational motion of the centre of mass of rod OD one has

$$(F_{r1} - F_{r2} - m_1 g \sin\theta)\, \boldsymbol{e_r} + (F_{\theta 1} - F_{\theta 2} - m_1 g \cos\theta)\, \boldsymbol{e_\theta} = m_1 \left(-h_1 \ddot{\theta} \boldsymbol{e_r} - h_1 \dot{\theta}^2 \boldsymbol{e_\theta}\right)$$

and by substitution, using equations (2) and (3),

$$F_{r1} = -\left(m_1 h_1 + m_2 h_2\right) \ddot{\theta} + \left(m_1 + m_2\right) g \sin\theta, \tag{4}$$

$$F_{\theta 1} = -\left(m_1 h_1 + m_2 h_2\right) \dot{\theta}^2 + \left(m_1 + m_2\right) g \cos\theta.$$

[6 marks]

e) Considering the motion about the centre of mass of rod OD according to $\frac{d\boldsymbol{H_1}}{dt} = \boldsymbol{N_1}$, where

$$H_1 = I_1 \dot{\theta},$$

and

$$N_1 = h_1 F_{r1} + (h_2 - h_1) F_{r2} - T_c = h_1 \left(F_{r1} - F_{r2}\right) + h_2 F_{r2} - T_c,$$

gives

$$I_1 \ddot{\theta} = h_1 \left(F_{r1} - F_{r2}\right) + h_2 F_{r2} - T_c.$$

By substitution, using equations (4) and (2), the first equation of motion is

$$\left(m_1 h_1^2 + m_2 h_2^2 + I_1\right) \ddot{\theta} - (m_1 h_1 + m_2 h_2)\, g \sin\theta = -T_c.$$

Considering the motion about the centre of mass of rod AB according to $\frac{d\boldsymbol{H_2}}{dt} = \boldsymbol{N_2}$, where $H_2 = I_2\left(\dot{\theta} + \dot{\phi}\right)$ and $N_2 = T_c$, gives the second equation of motion

$$I_2\left(\ddot{\theta} + \ddot{\phi}\right) = T_c.$$

[6 marks]

Answers